

Finley Desai

Palo Alto, CA | fwdesai@stanford.edu | [linkedin.com/in/finleydesai](https://www.linkedin.com/in/finleydesai) | [fwdesai.github.io](https://github.com/fwdesai)

Education

Stanford University

Palo Alto, CA

B.S. / M.S. Electrical Engineering, Coterminous Program – continuously enrolled through Jun 2028

M.S. Electrical Engineering – Expected Jun 2028 (graduate coursework beginning Fall 2026)

B.S. Electrical Engineering, Minor in Computer Science – Expected Jun 2027 | GPA: 3.99/4.0

Selected Coursework: Digital System Design, Circuits I-II, Signals and Systems I-II, Computer Systems, Machine Learning, Feedback Control Design

Technical Experience and Projects

Stanford Tambe Lab

Palo Alto, CA

Research Intern (Part-Time), Robotics Chip-on-Board Module

Jun 2026 – Present

- Designing a ~16-layer chip-on-board module in Altium for on-robot inference: board-level schematic through layout, integrating ~50 ICs including wire-bonded bare die.
- Developing the controlled-impedance stackup, microvia/blind-via architecture, and DDR3-speed routing constraints, using pre-layout signal-integrity simulation in MATLAB Signal Integrity Toolbox to close impedance and timing budgets before fabrication.
- Created Altium library footprints and bond-pad land patterns for the custom die, drawing die-attach and wire-bond geometry directly in layout.

Stanford Accelerate Lab

Palo Alto, CA

Research Intern, Hardware Accelerator Test PCB ([GitHub](#))

May 2026 – Jun 2026

- Designed a board-level bring-up and test PCB in KiCad for a custom accelerator ASIC, integrating JTAG and peripheral interfaces and closing clock routing for operation to 600 MHz.

Stanford Magnetic Resonance Systems Research Lab

Palo Alto, CA

Research Intern

Jun 2025 – Jan 2026

Custom High-Res Wireless Quadrature Digitizer PCB for MRI Radar Prototyping ([Conference Paper](#) | [GitHub](#))

- Designed a dual-channel quadrature digitizer with high-resolution ADCs and a custom analog front-end for continuous-wave radar vital signs monitoring in MRI; measured an ~80 dB improvement in dynamic range over SDR-based acquisition.
- Took the board from schematic and multilayer layout through hand assembly and bring-up, validating the analog signal chain with oscilloscopes, spectrum analyzers, signal generators, and multimeters.
- Developed MicroPython/PIO firmware on a Raspberry Pi Pico W for synchronized acquisition and Bluetooth streaming.
- Analyzed radar signal data using singular spectrum analysis to isolate respiratory and cardiac motion.

Stanford Molecular Imaging Instrumentation Laboratory

Palo Alto, CA

Research Intern

Mar 2024 – Jun 2025

- Developed and ran custom Monte Carlo physics simulations, analyzing large datasets to design and validate inter-crystal scatter correction and event-positioning algorithms for PET imaging.
- Two conference papers accepted for presentation at IEEE Nuclear Science Symposium & Medical Imaging Conference (NSS/MIC), first author on one; abstracts published in conference proceedings: [Event Positioning](#), [Inter-Crystal Scattering](#).

FPGA Digital Systems Projects – Verilog

- Designed and implemented FPGA digital systems in Verilog, including a bike-light controller and a polyphonic music player with harmonics and envelope modulation; validated each with simulation testbenches.

Additional Experience

American Industrial Partners

New York, NY

Operations Intern

Jun 2026 – Present

- Built an inventory-aging model for pulp woodyards estimating pine-chemical losses; presented to mill managers.
- Standardized maximum-sustainable-rate reporting across mills; built supporting financial models.

Technical Skills

Languages/Tools: Verilog, C, C++, Python, MATLAB, Altium Designer, KiCad, LTspice, Git, CMake, Linux

Hardware & Lab: Board-level design/layout/assembly, bring-up & debug, scopes, logic/spectrum analyzers, multimeters, SI analysis